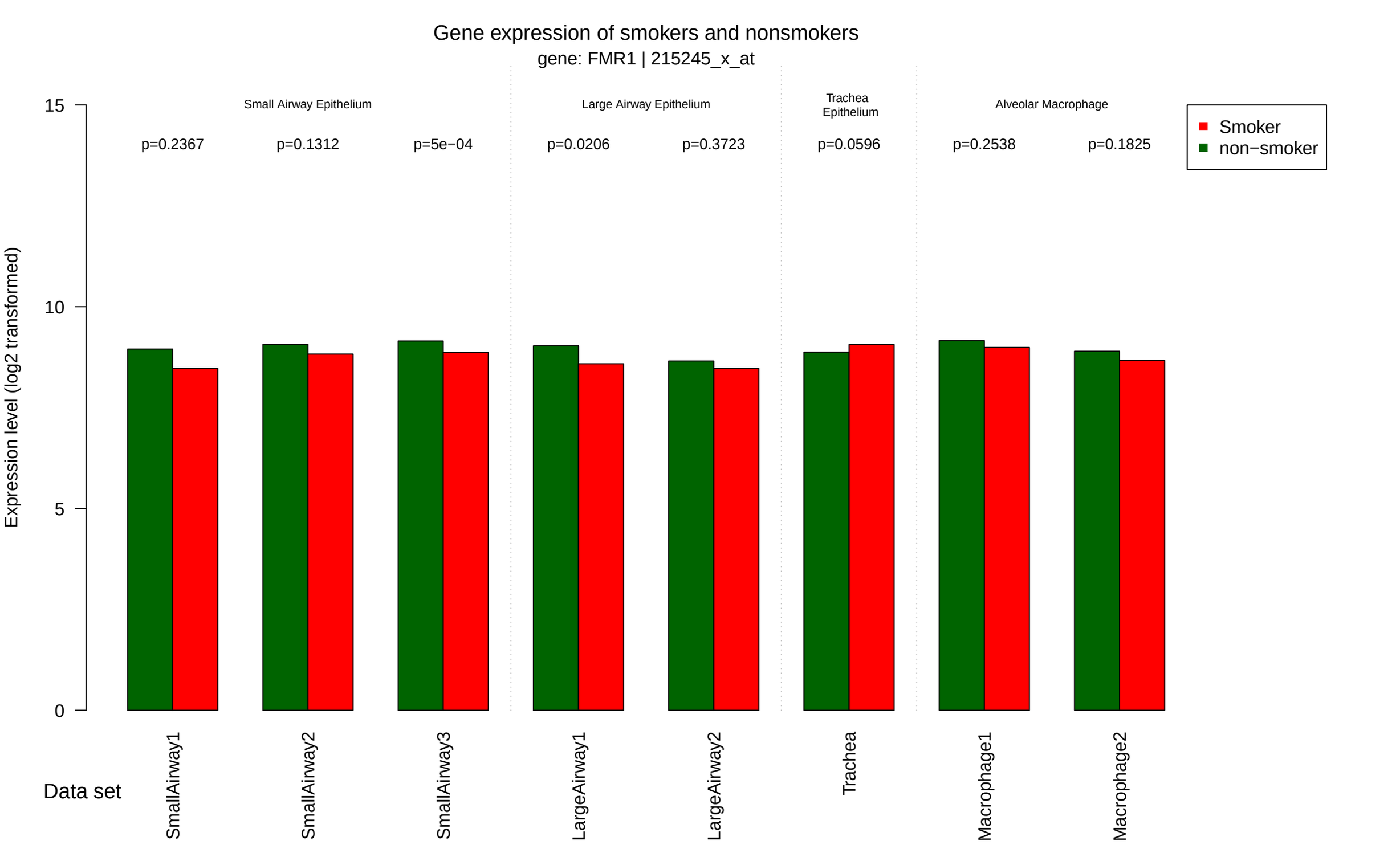
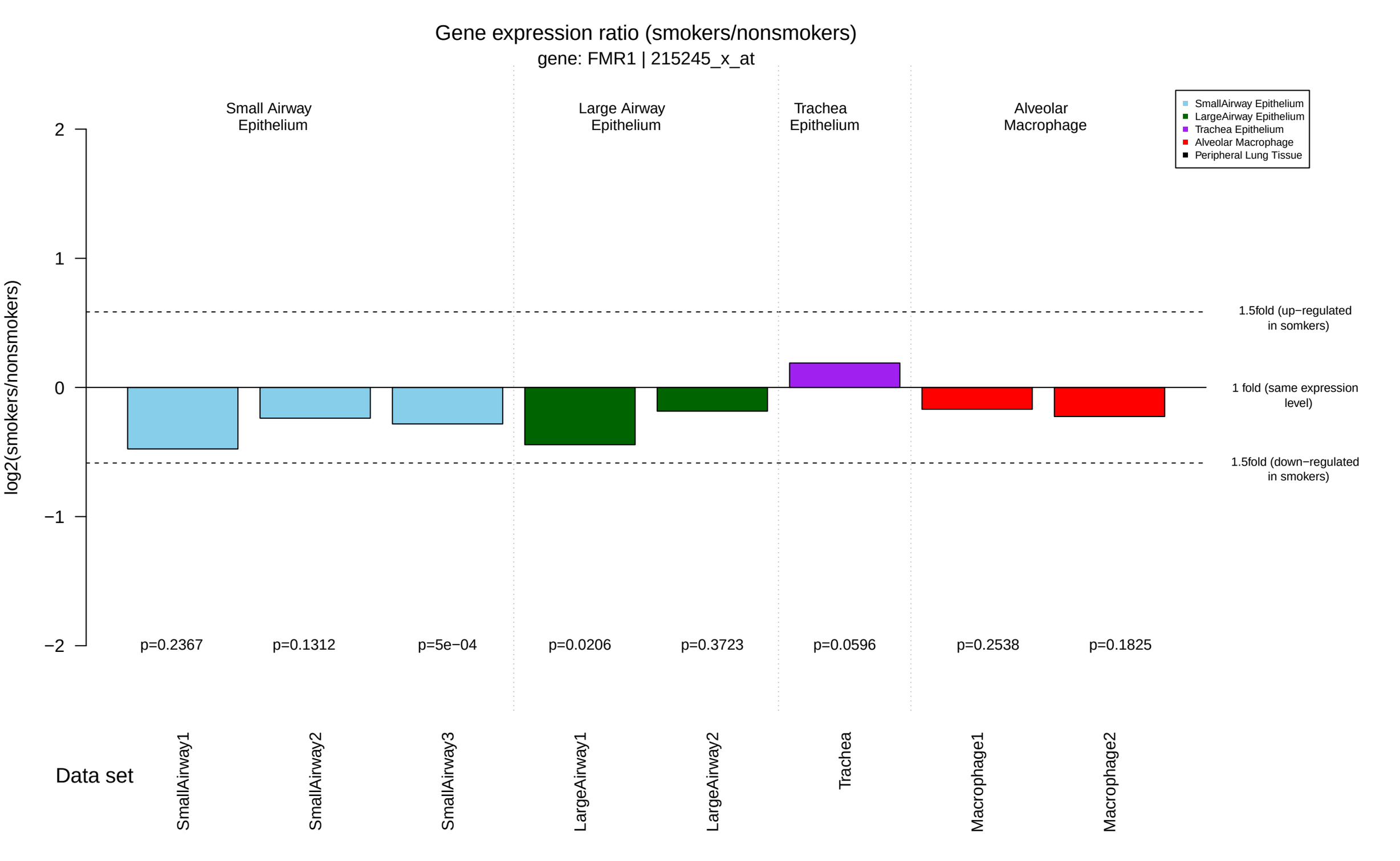
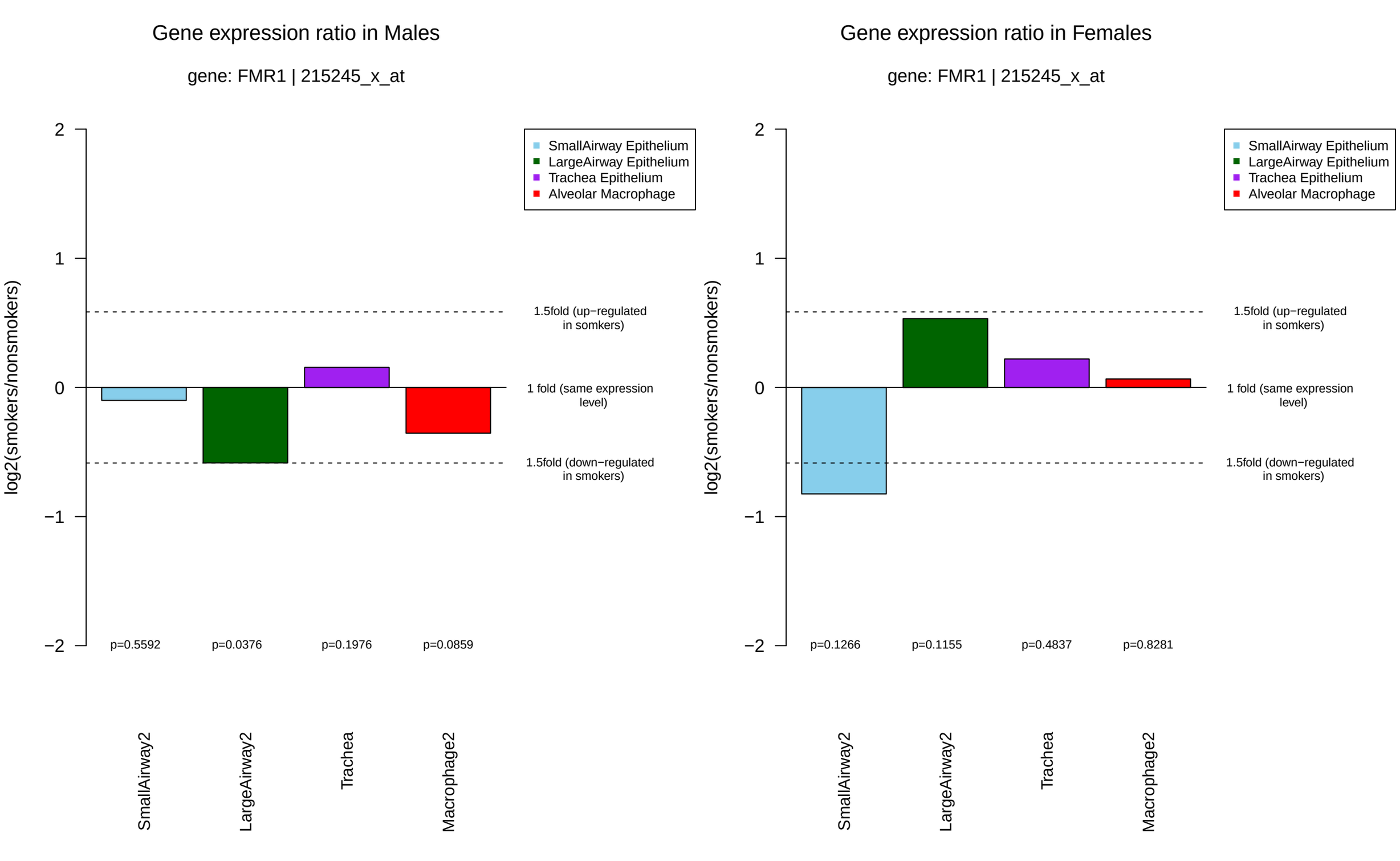
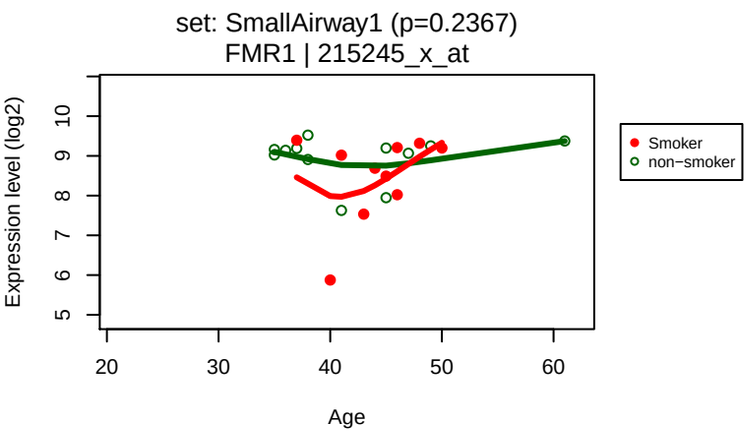
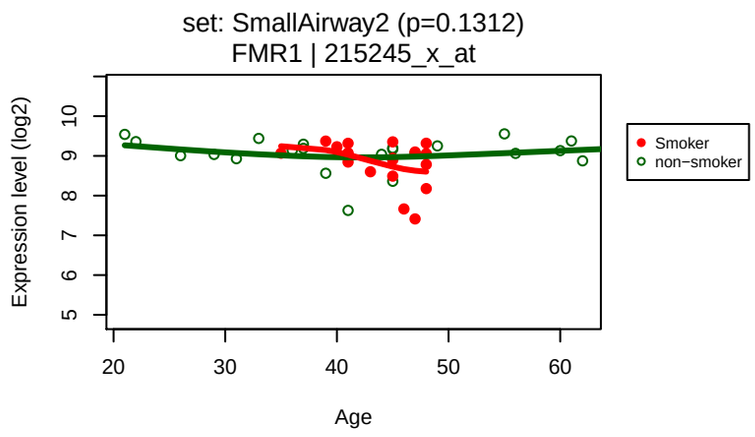
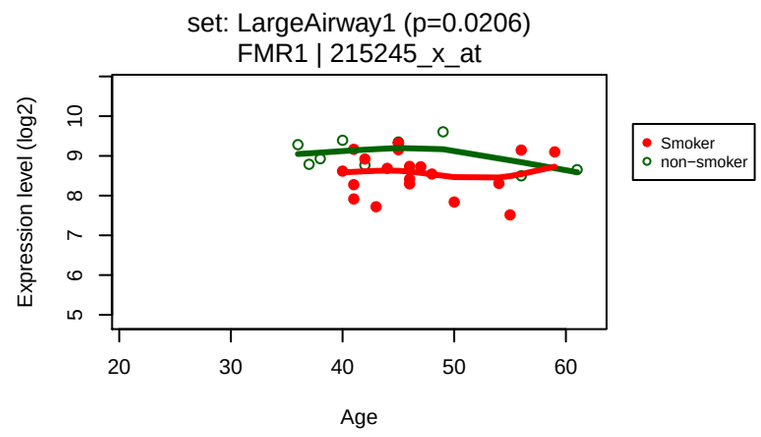

Gene expression of smokers and nonsmokers
gene: FMR1 | 215245_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2367
p=0.1312
p=5e−04
p=0.0206
p=0.3723
p=0.0596
p=0.2538
p=0.1825
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

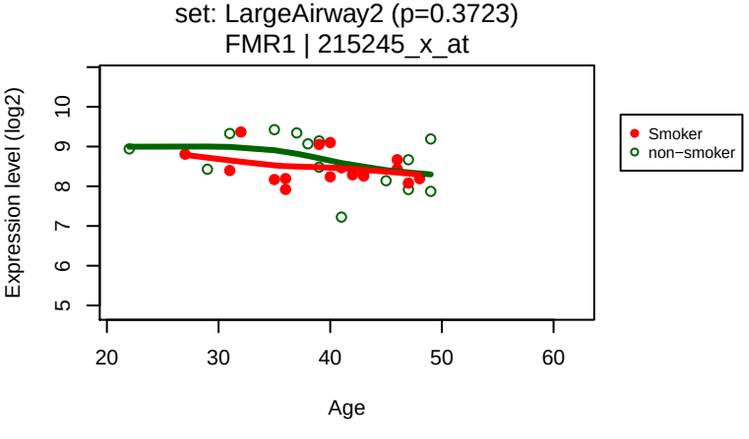

Gene expression ratio (smokers/nonsmokers)
gene: FMR1 | 215245_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2367
p=0.1312
p=5e−04
p=0.0206
p=0.3723
p=0.0596
p=0.2538
p=0.1825
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

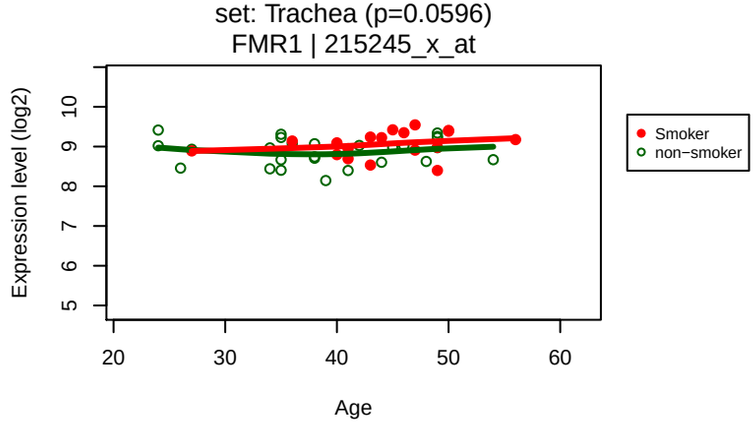

Gene expression ratio in Males
gene: FMR1 | 215245_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5592
p=0.0376
p=0.1976
p=0.0859
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FMR1 | 215245_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1266
p=0.1155
p=0.4837
p=0.8281
SmallAirway2
LargeAirway2
Trachea
Macrophage2

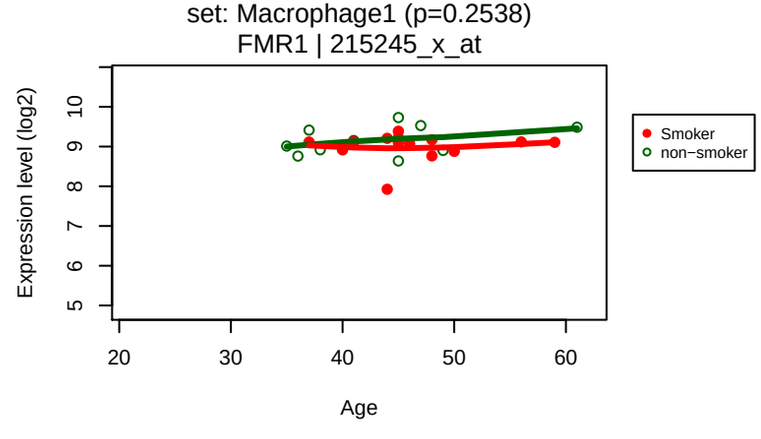

set: SmallAirway1 (p=0.2367)
FMR1 | 215245_x_at
Expression level (log2)
Age
Smoker
non−smoker

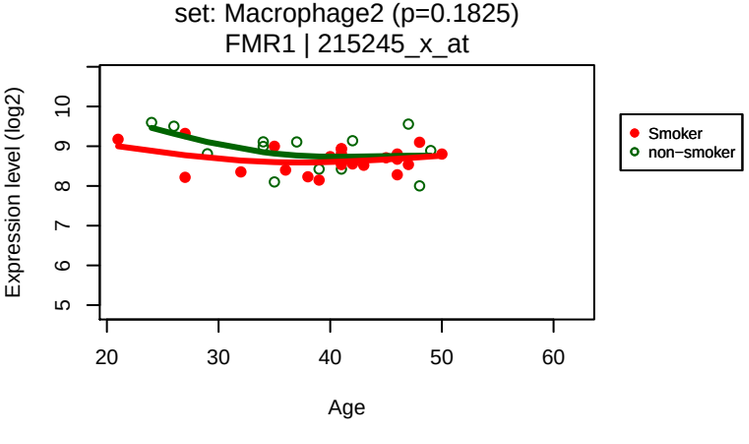

set: SmallAirway2 (p=0.1312)
FMR1 | 215245_x_at
Expression level (log2)
Age
Smoker
non−smoker

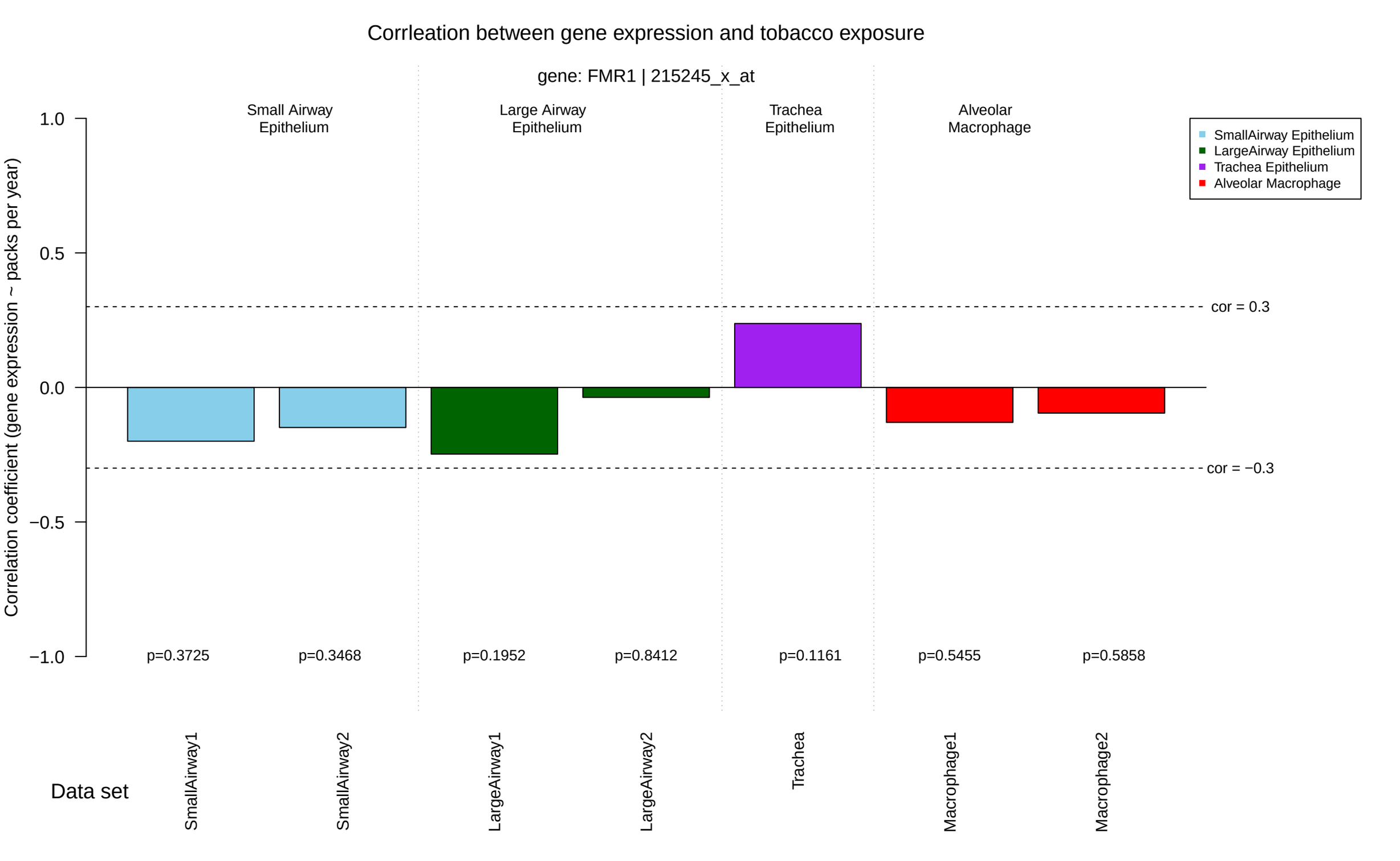

set: LargeAirway1 (p=0.0206)
FMR1 | 215245_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3723)
FMR1 | 215245_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0596)
FMR1 | 215245_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2538)
FMR1 | 215245_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1825)
FMR1 | 215245_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FMR1 | 215245_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3725
p=0.3468
p=0.1952
p=0.8412
p=0.1161
p=0.5455
p=0.5858
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2